

Naveen Jindal School of Management

Business Analytics and Artificial Intelligence (BS)

Bachelor of Science in Business Analytics and Artificial Intelligence

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Ashiq Ali, Alain Bensoussan, Gary Bolton, Metin Çakanyildirim, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Dorothée Honhon, Kyle Hyndman, Varghese S. Jacob, Sanjay Jain, Ganesh Janakiraman, Elena Katok, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Stanley Liebowitz, Zhiang (John) Lin, Sumit K. Majumdar, Stanimir Markov, Amit Mehra, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Özalp Özer, Mike W. Peng, Hasan Pirkul, Cuili Qian, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Suresh P. Sethi, Kathryn E. Steckle, Riki Takeuchi, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

Associate Professors: Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Michael Hasler, Bin Hu, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Maria Loumiotis, Livia Markóczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, H. Dennis Park, Anyan Qi, Young U. Ryu, Serdar Simsek, Harpreet Singh, Upender Subramanian, Shaojie Tang, Shouqiang Wang, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Yuan Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

Assistant Professors: Khai Chiong, Rafael Copat, Soraya Fatehi, Andrew Frazelle, Ying Huang, Joonhwi Joo, Sora Jun, Jason Kautz, Tongil Kim, Sheen Levine, Christopher Mace, Samir Mamadehussene, Jean-Marie Meier, Radha Mookerjee, Jedson Pinto, Ignacio Rios Uribe, Alejandro Rivera Mesias, Simon Siegenthaler, Kirti Sinha, Xiaoxiao Tang, Shervin Tehrani, Ashwin Venkataraman, Christian Von Drathen, Guihua Wang, Hongchang Wang, Pingle Wang, Junfeng Wu, Steven Xiao, Yingjie Zhang

Associate Professors Emeriti: J. Richard Harrison, Jane Salk, David J. Springate

Clinical Professors: John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Paul Convery, Howard Dover, John Gamino, Randall S. Guttery, William Hefley, Robert Hicks, Marilyn Kaplan, Sonia Leach, Peter Lewin, Jeffrey Manzi, John McCracken, Diane McNulty, Divakar Rajamani, Daniel Rajaratnam, Kannan Ramanathan, Mark Thouin, McClain Watson, Jeff Weekley, Habte Woldu, Fang Wu

Clinical Associate Professors: Shawn Alborz, Dawn Owens, Carolyn Reichert, Avanti P. Sethi, Ramesh Subramoniam, Aysegul Toptal, David Widdifield

Clinical Assistant Professors: Athena Alimirzaei, Christina (Krysta) Betanzos, Moran Blueshtein, Jerome Gafford, Jeffery (Jeff) Hicks, Revansiddha Khanapure, Kristen Lawson, Liping Ma, Ravi Narayan, Parneet Pahwa, Jason Parker, Nassim Sohaee

Professors of Instruction: Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Matt Polze, Luell (Lou) Thompson

Associate Professors of Instruction: Vivek Arora, Judd Bradbury, Monica E. Brussolo, Amal El-Ashmawi, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Kathryn Lookadoo, Sarah Moore, Hubert Zydorek, Mohammad aseri Taheri

Assistant Professors of Instruction: Negin Enayaty Ahangar, Joseph Mauriello, Victoria McCrady, Rasoul Ramezani, Gaurav Shekhar

Professors of Practice: Gregory Ballew, Tiffany A. Bortz, Ranavir Bose, Alexander Edsel, Rajiv Shah

Associate Professors of Practice: Richard Bowen, Jackie Kimzey, David Parks, Margaret Smallwood, Steven Solcher, Kathy Zolton

Assistant Professors of Practice: Khatereh Ahadi, Steven Haynes, Abu Naser Islam, Edward Meda, Paul Nichols, Timothy Stephens, Guido Tirone, Robert Wright

I. Core Curriculum Requirements: 42 semester credit hours¹

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 1325](#) Applied Calculus I^{2, 3, 4, 5}

Or select any 3 semester credit hours from [Mathematics Core](#) courses⁴ (see advisor)

Life and Physical Sciences: 6 semester credit hours

Select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor and degree requirements)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours⁶

Choose one of the following:

[BA 1310](#) Making Choices in Free Market Systems^{2, 3}

[BA 1320](#) Business in a Global World^{2, 3}

[ECON 2301](#) Principles of Macroeconomics^{2, 3}

[ECON 2302](#) Principles of Microeconomics^{2, 3}

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours⁶

Choose two of the following:

[MATH 1326](#) Applied Calculus II^{2, 3, 4}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3}

[BA 1320](#) Business in a Global World^{2, 3}

[ECON 2301](#) Principles of Macroeconomics^{2, 3} _{_}

[ECON 2302](#) Principles of Microeconomics^{2, 3} _{_}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 69 semester credit hours

Students must meet all course prerequisites.

Major Preparatory Courses: 18 semester credit hours beyond Core Curriculum

[ACCT 2301](#) Introductory Financial Accounting

[ACCT 2302](#) Introductory Management Accounting² _{_}

[BLAW 2301](#) Business and Public Law² _{_}

[ITSS 3311](#) Introduction to Programming⁷ _{_}

[MATH 1325](#) Applied Calculus I^{2, 3, 4, 5} _{_ _}

[MATH 1326](#) Applied Calculus II^{2, 3, 4, 8} _{_ _}

or [OPRE 3340](#) Advanced Business Quantitative Methods⁸ _{_}

[OPRE 3333](#) Quantitative Business Analysis^{2, 9} _{_}

or [MATH 2333](#) Matrices, Vectors, and Data^{2, 9} _{_}

[OPRE 3360](#) Managerial Methods in Decision Making Under Uncertainty

or [STAT 3360](#) Probability and Statistics for Management and Economics

Choose two of the following:⁶ _{_}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3} _{_}

[BA 1320](#) Business in a Global World^{2, 3} _{_}

[ECON 2301](#) Principles of Macroeconomics^{2, 3} _{_}

[ECON 2302](#) Principles of Microeconomics^{2, 3} _{_}

Major Core Courses: 24 semester credit hours

[BCOM 1300](#) Professional Communication⁹

or [BCOM 3300](#) Business Communication⁹

[BCOM 4300](#) Advanced Professional Communication

[FIN 3320](#) Business Finance

[IMS 3310](#) International Business

[ITSS 3300](#) Information Technology for Business

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

or [OBHR 3330](#) Introduction to Human Resource Management

[OPRE 3310](#) Supply Chain and Operations Management

Major Related Courses: 27 semester credit hours

[BUAN 4320](#) Database Fundamentals for Analytics

[BUAN 4373](#) Data Science for Business Applications

[BUAN 4350](#) Spreadsheet Modeling and Analytics

or [OPRE 4350](#) Spreadsheet Modeling and Analytics

[BUAN 4351](#) Foundations of Business Intelligence

or [ITSS 4351](#) Foundations of Business Intelligence

[BUAN 4353](#) Business Analytics

or [ITSS 4353](#) Business Analytics

[BUAN 4355](#) Data Visualization

or [ITSS 4355](#) Data Visualization

[BUAN 4381](#) Object Oriented Programming with Python

or [ITSS 4381](#) Object Oriented Programming with Python

[BUAN 4382](#) Applied Artificial Intelligence/Machine Learning

or [ITSS 4382](#) Applied Artificial Intelligence/Machine Learning

[BUAN 4395](#) Capstone Senior Project - Business Analytics

or [BPS 4395](#) Capstone Senior Project - Business

A practicum experience of at least 160 working hours is required:

[BUAN 4090](#) Business Analytics Internship

or [BA 4090](#) Management Internship¹⁰

Students are required to complete 100 hours of community engagement:

[BA 4095](#) Social Sector Engagement and Community Outreach Practicum¹¹

III. Elective Requirements: 9 semester credit hours

Guided Electives: 9 semester credit hours

The BS BUAN degree program gives students the opportunity to focus on a specific Concentration to obtain in-depth knowledge in a specific business area. Students will choose 9 semester credit hours from one of the following Concentrations:

A. Marketing Analytics Concentration

Students who pursue this concentration learn skills necessary for using data to gather deeper consumer insights, optimizing marketing objectives, and in general, achieving a higher return on investment. Students completing this concentration may pursue careers as Marketing and Research Analysts, Social Media and Digital Marketing Specialists.

[MKT 4330](#) Digital and Internet Marketing

[MKT 4331](#) Digital Prospecting

[MKT 4334](#) Social Media Marketing

[BUAN 4337](#) Marketing Analytics

or [MKT 4337](#) Marketing Analytics

B. Operations and Supply Chain Analytics Concentration

Students who pursue this concentration learn to apply analytical concepts in diverse areas in supply chain industry such as integrated supply chain management, project management and inventory management to name a few. Students completing this concentration may pursue careers as Supply Chain Analysts, Project Managers and Product Analysts.

[OPRE 3320](#) Integrated Supply Chain Management

[OPRE 4320](#) Integrated SCM Information Systems

[OPRE 4330](#) Global Logistics and Inventory Management

[BUAN 4357](#) Supply Chain Analytics, AI, and Advanced Solutions

or [OPRE 4357](#) Supply Chain Analytics, AI, and Advanced Solutions

C. Finance and Risk Analytics Concentration

Students who choose this concentration will learn both the qualitative and quantitative tools necessary to enter one of the many different areas within the finance risk management and insurance industries. Students completing this concentration may pursue careers as Financial Analysts, Quantitative Risk Analysts and Credit Risk Model Developers.

[FIN 3390](#) Introduction to Financial Modeling

[FIN 4300](#) Investment Management

[FIN 4338](#) Foundations of Risk Analytics and Applications

[FIN 4346](#) Applied Machine Learning in Finance, Insurance, and Real Estate

D. Information Technology Concentration

Students who choose this concentration will learn key fundamentals of how businesses design and use information technology to optimize their business processes. Combined with business analytics knowledge students completing this concentration may pursue careers as Business Intelligence Analysts, IT Analysts and Systems Analysts.

[ITSS 4330](#) Systems Analysis and Design

[ITSS 4340](#) Enterprise Resource Planning

[ITSS 4344](#) CRM using Salesforce

[ITSS 4356](#) Data Governance

[ITSS 4360](#) Network and Information Security

[ITSS 4371](#) Cloud Computing with AWS

E. Data Science Concentration

Students who choose this concentration will advance their knowledge into domains of advanced applied machine learning, web analytics, artificial intelligence, and big data analytics. Students may

pursue job titles such as Data Scientist, Data Mining Engineer, Data Analyst, Decision Scientist, Machine Learning Scientist, Data Manager and Data Architect.

[BUAN 4352](#) Introduction to Web Analytics

or [ITSS 4352](#) Introduction to Web Analytics

[BUAN 4354](#) Advanced Big Data Analytics

or [ITSS 4354](#) Advanced Big Data Analytics

[BUAN 4383](#) Advanced Applied Artificial Intelligence/Machine Learning

or [ITSS 4383](#) Advanced Applied Artificial Intelligence/Machine Learning

F. General Concentration

Students who pursue this concentration have the flexibility to apply their analytical knowledge and skills to diverse areas not limited to business domains of Marketing, Finance, Supply Chain, or IT, but also in areas of Litigation, Social Sciences, Sports Analytics and many more. Students pursuing this concentration and any of the above, may pursue the job title of Business Analysts.

Any combination of approved upper-level guided elective courses listed in Tracks A, B, C, D, or E may be used to satisfy the required 9 hours of guided elective credit.

All prerequisites must be met.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
2. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
3. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
4. MATH 1325 is required for all JSOM undergraduate students. Students may elect to substitute MATH 2413 or MATH 2417 for MATH 1325. Students may elect to substitute MATH 2414 or MATH 2419 for MATH 1326.
5. To make timely degree progress, students should complete MATH 1325 or its equivalent by the end of their first semester at UT Dallas. Students who will not meet this requirement should contact their academic advisor to discuss their degree timeline.
6. Certain courses listed are prerequisites for major core, major concentration, or major related

courses. Choose accordingly.

7. Students who have completed CS 1436 with a C or better may replace ITSS 3311 with an additional BUAN upper-division or ITSS upper-division course (excluding ITSS 4301).

8. Students may substitute OPRE 3340, but MATH 1326 is recommended for efficiency. Students who complete OPRE 3340 instead of MATH 1326 or MATH 2414 or MATH 2419 will need an 090 Component Area Option Core course.

9. JSOM first-time-in-college freshmen are required to take BCOM 1300 in their first semester. Transfer students and students new to JSOM are required to take BCOM 3300 in their first semester.

10. Students who are employed in full-time jobs may fulfill their internship requirement with their current employer as a special project with preapproval from their program director.

11. Students may fulfill the community engagement requirement with BA 4095, BPS 4396, ENTP 4340, IMS 4335, or MKT 4360. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement.

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